

Sujith Julakanti

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EDUCATION

Texas A&M University, College Station, TX (GPA:4/4)

United States

Master of Science in Data Science \$1000 Scholarship awardee

Aug 2024 – May 2026

National Institute of Technology Puducherry (GPA: 3.6/4)

Puducherry, India

Bachelor of Technology in Electronics and Communication Engineering

Aug 2019 – Apr 2023

SKILLS

Programming: Python, C++, C, SQL, HTML, CSS, ReactJS, MATLAB

Frameworks & Tools: PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Matplotlib, LangGraph, Flask, LoRA Finetuning, PEFT, Postman, Jenkins, Git, Docker

Technologies & Platforms: Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Computer Vision, Azure, AWS, Kubernetes, Grafana, Tableau, Power BI, Elasticsearch, MySQL, MongoDB, Microsoft Office Suite, Excel, Automation Solutions, Agile

Competencies: Deep Learning (Transformers, CNNs, RNNs, FFNNs), Natural Language Processing (NLP), Data Structures, Data Analysis, Statistical Modeling, Mathematical Modeling, Data Visualization, Statistics

WORK EXPERIENCE

Texas A&M University

College Station, Texas

Graduate Student Assistant

Jan 2025-May 2026

- Implemented a patch-wise Stable Diffusion inpainting pipeline to seamlessly remove and replace detected objects (e.g., vehicles) in images using YOLO-based detection and mask generation.
- Spearheaded the development of a Graduate Portal using Django to collect instructor feedback, applying LLMs to extract insights for academic evaluation.

ASEA BROWN BOVERI (ABB)

Bangalore, India

MLOps Engineer

Jul 2023 – Jul 2024

- Engineered a customer-specific interaction chatbot using LangGraph to build conversational flows, combined with LLMs for intent recognition and CRM systems for context-aware responses, improving customer engagement and query resolution efficiency.
- Automated cloud infrastructure provisioning by deploying 50+ resources using Terraform, reducing manual effort by 90% and deployment time by 75%.

ASEA BROWN BOVERI (ABB)

Bangalore, India

Software Engineering Intern

Jan 2023 – Jun 2023

- Built a system health monitoring tool using the EFK stack, reducing downtime by 30% across monitored systems.
- Created APIs for monitoring machine learning model health, processing over 5,000 model metrics per day and enabling a 20% reduction in model maintenance time.

PUBLICATIONS

- Journal Paper:** Published a research paper titled Computational Analysis of Various Techniques for Classification of Liver Disease. Citation J. Sujith et al 2023 J. Phys.: Conf. Ser. 2466 .

PROJECTS

- TreeClear – Satellite Image Enhancer for Residential Visibility:** Designed a vision pipeline integrating YOLOv11 for object detection, SAM for instance segmentation, and Stable Diffusion for context-aware inpainting of occluded regions in satellite imagery.
- Oral Cancer Detection Using Signal Processing Algorithms And Deep Learning Techniques:** Formulated a feature extraction framework leveraging MFCC and STFT for audio-based Oral Cancer detection. Enhanced accuracy with ResNet and InceptionV3 models, reaching 94% precision and 93% recall. Explored GMM to refine clustering and performance consistency.
- AssistIQ:** Launched an interactive multimodal LLM agent widget with 1,000+ downloads, seamlessly embedded into laptops to process text, voice, and image inputs for daily productivity tasks such as scheduling, document summarization, and contextual search. Leveraged on-device APIs, custom inference pipelines to enable real-time, low-latency desktop performance.